

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56e)

Metric

$$ds^2 = \left(\frac{12 \kappa_0 m_0 q_0^4 r u^{\frac{5}{3}} + 18 a^2 r - (3 \kappa_0^2 q_0^6 u^2 e^x - 2 (\kappa_0^2 q_0^6 e^{(2x)} - 6 a \kappa_0 q_0^3 e^x) r u) u^{\frac{1}{3}}}{6 \kappa_0 q_0^4 r^2 u^{\frac{5}{3}}} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 e^x dx \otimes dx + r^2 e^x dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (-1) du$$

$$n = \left(\frac{\kappa_0 q_0^2 e^{(2x)}}{6 r u^{\frac{1}{3}}} - \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4 r^2} + \frac{m_0}{r} - \frac{a e^x}{q_0 r u^{\frac{1}{3}}} + \frac{3 a^2}{2 \kappa_0 q_0^4 r u^{\frac{5}{3}}} \right) du + (-1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r e^{(\frac{1}{2} x)} \right) dx + \left(\frac{1}{2} i \sqrt{2} r e^{(\frac{1}{2} x)} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r e^{(\frac{1}{2} x)} \right) dx + \left(-\frac{1}{2} i \sqrt{2} r e^{(\frac{1}{2} x)} \right) dy$$

Contravariant components

$$l = (1) dr$$

$$n = (1) \, du + \left(\frac{\kappa_0 q_0^2 e^{(2x)}}{6 r u^{\frac{1}{3}}} - \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4 r^2} + \frac{m_0}{r} - \frac{a e^x}{q_0 r u^{\frac{1}{3}}} + \frac{3 a^2}{2 \kappa_0 q_0^4 r u^{\frac{5}{3}}} \right) dr$$

$$m = \left(\frac{\sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \right) dx + \left(\frac{i \sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \right) dx + \left(-\frac{i \sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \right) dy$$

Directional Derivatives

$$D = \partial_r$$

$$\Delta = \partial_u + \frac{\kappa_0 q_0^2 e^{(2x)}}{6 r u^{\frac{1}{3}}} - \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4 r^2} + \frac{m_0}{r} - \frac{a e^x}{q_0 r u^{\frac{1}{3}}} + \frac{3 a^2}{2 \kappa_0 q_0^4 r u^{\frac{5}{3}}} \partial_r$$

$$\delta = \frac{\sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \partial_x + \frac{i \sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \partial_x - \frac{i \sqrt{2} e^{(-\frac{1}{2} x)}}{2 r} \partial_y$$

Spin Coefficients

$$\rho = -\frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\kappa_0 q_0^2 e^{(2\,x)}}{6\,r^2 u^{\frac{1}{3}}} - \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4\,r^3} + \frac{m_0}{r^2} - \frac{a e^x}{q_0 r^2 u^{\frac{1}{3}}} + \frac{3\,a^2}{2\,\kappa_0 q_0^4 r^2 u^{\frac{5}{3}}}$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{2}\kappa_0 q_0^2 e^{\left(\frac{3}{2}\,x\right)}}{6\,r^2 u^{\frac{1}{3}}} + \frac{\sqrt{2}\kappa_0 q_0^2 u^{\frac{2}{3}} e^{\left(\frac{1}{2}\,x\right)}}{8\,r^3} + \frac{\sqrt{2}a e^{\left(\frac{1}{2}\,x\right)}}{2\,q_0 r^2 u^{\frac{1}{3}}}$$

$$\alpha = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,x\right)}}{8\,r}$$

$$\beta = \frac{\sqrt{2}e^{\left(-\frac{1}{2}\,x\right)}}{8\,r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{\kappa_0 q_0^2 e^{(2\,x)}}{12\,r^2 u^{\frac{1}{3}}} - \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4\,r^3} + \frac{m_0}{2\,r^2} - \frac{a e^x}{2\,q_0 r^2 u^{\frac{1}{3}}} + \frac{3\,a^2}{4\,\kappa_0 q_0^4 r^2 u^{\frac{5}{3}}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{4 r^4}$$

$$\Phi_{12} = \frac{\sqrt{2} \kappa_0 q_0^2 e^{\left(\frac{3}{2} x\right)}}{12 r^3 u^{\frac{1}{3}}} - \frac{\sqrt{2} \kappa_0 q_0^2 u^{\frac{2}{3}} e^{\left(\frac{1}{2} x\right)}}{8 r^4} - \frac{\sqrt{2} a e^{\left(\frac{1}{2} x\right)}}{4 q_0 r^3 u^{\frac{1}{3}}}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2} \kappa_0 q_0^2 e^{\left(\frac{3}{2} x\right)}}{12 r^3 u^{\frac{1}{3}}} - \frac{\sqrt{2} \kappa_0 q_0^2 u^{\frac{2}{3}} e^{\left(\frac{1}{2} x\right)}}{8 r^4} - \frac{\sqrt{2} a e^{\left(\frac{1}{2} x\right)}}{4 q_0 r^3 u^{\frac{1}{3}}}$$

$$\Phi_{22} = \frac{\kappa_0 q_0^2 e^{(2 x)}}{18 r^2 u^{\frac{4}{3}}} - \frac{\kappa_0 q_0^2 e^x}{6 r^3 u^{\frac{1}{3}}} + \frac{\kappa_0 q_0^2 u^{\frac{2}{3}}}{8 r^4} - \frac{a e^x}{3 q_0 r^2 u^{\frac{4}{3}}} + \frac{a}{2 q_0 r^3 u^{\frac{1}{3}}} + \frac{5 a^2}{2 \kappa_0 q_0^4 r^2 u^{\frac{8}{3}}}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\kappa_0 q_0^2 e^{(2x)}}{6 r^3 u^{\frac{1}{3}}} + \frac{\kappa_0 q_0^2 u^{\frac{2}{3}} e^x}{2 r^4} - \frac{m_0}{r^3} + \frac{a e^x}{q_0 r^3 u^{\frac{1}{3}}} - \frac{3 a^2}{2 \kappa_0 q_0^4 r^3 u^{\frac{5}{3}}}$$

$$\Psi_3 = \frac{\sqrt{2} \kappa_0 q_0^2 e^{(\frac{3}{2} x)}}{4 r^3 u^{\frac{1}{3}}} - \frac{\sqrt{2} \kappa_0 q_0^2 u^{\frac{2}{3}} e^{(\frac{1}{2} x)}}{4 r^4} - \frac{3 \sqrt{2} a e^{(\frac{1}{2} x)}}{4 q_0 r^3 u^{\frac{1}{3}}}$$

$$\Psi_4 = -\frac{\kappa_0 q_0^2 e^x}{6 r^3 u^{\frac{1}{3}}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"